

Adrenal Crisis



Section I: Scenario Demographics

Scenario Title:	Adrenal Crisis		
Date of Development:	02/03/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a rare presentation that requires important critical care steps in its management
CRM Objectives:	1) Communicate clearly with team members during complicated resuscitation. 2) Lead team members effectively through the management of a critically ill patient.
Medical Objectives:	1) Simultaneously resuscitate and initiate investigations for the critically ill patient with vague symptoms. 2) Recognize non-fluid responsive shock. 3) Recognize and appropriately treat the hyperkalemia and hypoglycemia associated with an adrenal crisis. 4) Initiate appropriate steroid treatment for an adrenal crisis.

Case Summary: Brief Summary of Case Progression and Major Events
A 46-year-old female presents to the ED complaining of fatigue, anorexia, and weight loss over the last two weeks. She had the "stomach flu" a couple weeks ago and thought she was getting over it. But now she feels very weak and seems to be vomiting again. On presentation, the patient will have mild hypothermia, hypoglycemia, and hypotension. The team will have to initiate fluid resuscitation and an initial workup. The patient's blood pressure won't respond to 4 L of IV fluids, forcing the residents to work through the differential diagnosis of shock. Eventually, they will receive critical VBG results that indicate a mild metabolic acidosis, hyperkalemia, and hyponatremia. The team will need to treat the hyperkalemia and initiate hydrocortisone therapy.
References
Sharma A, Levy D. (2013). Thyroid and adrenal disorders. In J. Marx, R. Hockberger & R. Walls (Eds.), Rosen's emergency medicine - concepts and clinical practice. pp. (1689-92). Philadelphia, PA:Saunders.

Adrenal Crisis



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
<i>Select most important dimension(s)</i>			
Confederates Brief Description of Role			
None.			
B. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☐ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	5 min	Scenario:	15 min
		Debriefing:	20 min

Adrenal Crisis



Section V: Patient Data and Baseline State

A. Clinical Vignette: to Read Aloud at Beginning of Case

A 46-year-old female presents to the ED complaining of fatigue, anorexia, and weight loss over the last two weeks. She had the “stomach flu” a couple weeks ago and thought she was getting over it. But now she feels very weak and seems to be vomiting again. Her blood pressure is 80/40, so she was triaged straight to the resuscitation bay.

B. Patient Profile and History

Patient Name: Andrea Nealy	Age: 42	Weight: 65 kg
Gender: ☐ M ☒ F	Code Status: Full	

Chief Complaint: Vomiting

History of Presenting Illness: Had “stomach flu” two weeks ago. Thought she was getting better. But now, over the last few days, feeling weak, tired, and starting to vomit again.

Past Medical History:	Depression	Medications:	Celexa 20mg OD

Allergies: None.

Social History: Single mother of two (ages 10 and 12). **Smokes ½ ppd. Occasional EtOH.** No recreational drugs.

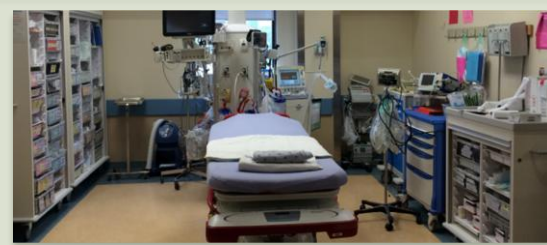
Family History: Nil.

Review of Systems:	CNS:	Feels tired and weak.		
	HEENT:	Nil.		
	CVS:	Nil.		
	RESP:	Nil.		
	GI:	Progressively vomiting more and more. No blood. No bile. Unable to maintain PO. No diarrhea. Minimal abdo pain.		
	GU:	No lower urinary tract symptoms.		
	MSK:	Nil.	INT:	Feels really chilled.

B. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 125/min	BP: 80/40	RR: 18/min
T: 35°C	Glucose: 2.6 mmol/L	O ₂ SAT: 98 % RA
General Status: Appears unwell. Looks pale and dehydrated.		
CNS:	PERLA, 2-3mm. A+Ox3, answering questions. No focal deficits.	
HEENT:	Nil acute.	
CVS:	Normal HS, no murmurs. Weak peripheral pulses.	
RESP:	GAEB. No adventitious.	
ABDO:	Soft, non-tender.	
GU:	Nil acute.	
MSK:	No hot joints.	SKIN: Pale. No rashes.

Adrenal Crisis



Section VI: Scenario Progression

Scenario States, Modifiers and Progression			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 125/min BP: 80/40 RR: 18/min O ₂ SAT: 98 % RA T: 35°C	A+Ox3. Looks pale, dehydrated, unwell.	<u>Learner Actions</u> - ☐ Monitors - ☐ IV access, bolus 2L - ☐ Order septic work-up, liver panel, VBG, lactate (broad) - ☐ Order urine or serum βHCG - ☐ Check sugar, give glucose - ☐ Administer antibiotics - ☐ Order CXR - ☐ Order urinalysis - ☐ Start vasopressors after IVF don't correct BP - ☐ Bedside U/S to assess for FF or PCE	<u>Modifiers</u> - Give 2L bolus → HR 115 - Give 4L bolus → HR 110 - No blood work ordered by 5 min → frequent PVC's <u>Triggers</u> - Fluids given, blood work ordered, glucose corrected, or 10 min. → 2. Blood work back - Capillary glucose not checked by 5 min. → 3. Seizure
2. Blood work back HR → 110 BP → 85/40	Clinically less dry. Still looks unwell.	<u>Learner Actions</u> - ☐ ID hyperkalemia, do ECG - ☐ Treat hyperkalemia: 1) Calcium gluconate 1g iv 2) D50 + 10 units insulin R 3) Ventolin 16 puffs 4) NaHCO ₃ 1 amp - ☐ ID hyponatremia and treat slowly - ☐ ID adrenal crisis as most likely diagnosis, treat with hydrocortisone 100mg iv	<u>Modifiers</u> - Give hydrocortisone → BP to 100/60 <u>Triggers</u> - Don't recognize adrenal crisis by 12 min. → 4. VF arrest - Don't treat hyperkalemia by 12 min. → 4. VF arrest - Correct potassium and give hydrocortisone → End Case
3. Seizure HR → 140 BP → unable RR → no effort O ₂ SAT → 90% despite FiO ₂	Patient having tonic clonic seizures.	<u>Learner Actions</u> - ☐ Check blood glucose - ☐ Correct glucose with D50	<u>Modifiers</u> - Patient will keep seizing until glucose given. <u>Triggers</u> - Glucose given → 2. Blood work back - Intubation without glucose check → 4. VF arrest
4. VF arrest Rhythm → VF HR → 180, no pulse BP → unable	Patient pulseless with VF rhythm x3 rounds.	<u>Learner Actions</u> - ☐ Good quality CPR - ☐ Shock VF - ☐ Epinephrine - ☐ Amiodarone - ☐ ±Intubation	<u>Triggers</u> - 15 min → End Case

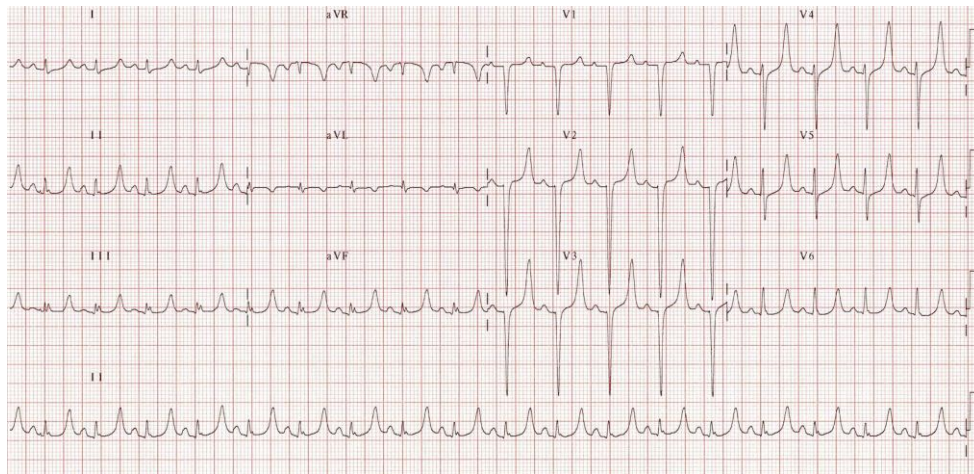
Adrenal Crisis



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 120	K: 7.5	Cl: 100	HCO ₃ : 17	BUN: 10	Cr: 35	Glu: 1.5
VBG	pH: 7.31	PCO ₂ : 32	PO ₂ : 45	HCO ₃ : 18	Lactate: 3.1	
WBC: 18.5		Hg: 140		Hct: 0.66		Plt: 550

Images (ECGs, CXRs, etc.)



ECG source: <http://lifeinthefastlane.com/ecg-library/basics/hyperkalaemia/>



CXR source: <https://radiopaedia.org/cases/normal-chest-radiograph-female-1>

Ultrasound Video Files (if applicable)

Cardiac Ultrasound – no PCE

FAST – no free fluid



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What was your original differential diagnosis for this critically ill patient? How did that change throughout the case? What acted as a trigger to your thought process? Did it feel uncomfortable to change your working diagnosis? 2) Did you find it challenging to trouble-shoot this patient's refractory shock? How do you feel to worked to problem-solve this as a team? 3) What are the laboratory abnormalities associated with a primary adrenal crisis? How are they different in secondary adrenal failure? Why? 4) Why is hydrocortisone the steroid used for treatment in acute adrenal crisis? What are other options?	
Key Moments	
1) Recognition that the patient is critically ill and not responding to IVF.	
2) Identification and treatment of hypoglycemia.	
3) Identification of possible adrenal crisis.	